Module 5 slides: Oral Absorption (worked examples)

Module 5 - Single Oral Administration · Lecture slides, worked examples · 16 Sep 2026

Name _____ Date _____

Example 1 An investigational new drug has a half-life of absorption of 45 minutes. What is its absorption rate constant?

Example 1 The same investigational drug has a half-life of elimination of 3 hours. What is its elimination rate constant?

Example 1 The following pharmacokinetic parameters were found for an investigational new drug: V_D is 22 L, the half-life of absorption is 45 minutes and the half-life of elimination is 3 hours. The drug is 85% bioavailable and follows a one-compartment model. When does the maximum concentration occur after a single 500-mg oral dose?

Example 1 For the same investigational drug, with V_D of 22 L, an absorption half-life of 45 minutes, an elimination half-life of 3 hours and 85% bioavailability, what maximum concentration is expected after a single 500-mg oral dose?

Example 1 A 500-mg oral dose of an investigational drug with V_D of 22 L, an absorption half-life of 45 minutes, an elimination half-life of 3 hours and 85% bioavailability produces a peak concentration of 12.17 mg/L. What peak concentration is expected if the dose is increased to 1000 mg?

Example 2 A single oral dose of an antibiotic was given to an adult 70 kg male patient. The drug fits a one-compartment open model with $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with C_p in mcg/mL and t in hours. What is the elimination half-life for this drug in this patient?

Example 2 For the same antibiotic, described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$ with t in hours, what is the half-life of absorption?

Example 2 A single oral dose of an antibiotic given to an adult 70 kg male patient is described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with C_p in mcg/mL and t in hours. Calculate t_{max}.

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Example 2 The antibiotic described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with concentration in mg/L and time in hours, was given as a 500-mg oral dose that is 87% bioavailable. What is the apparent volume of distribution?

Example 2 The antibiotic described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with C_p in mg/L and t in hours, was given as a single oral dose. What plasma concentration is expected 4 hours after the dose?